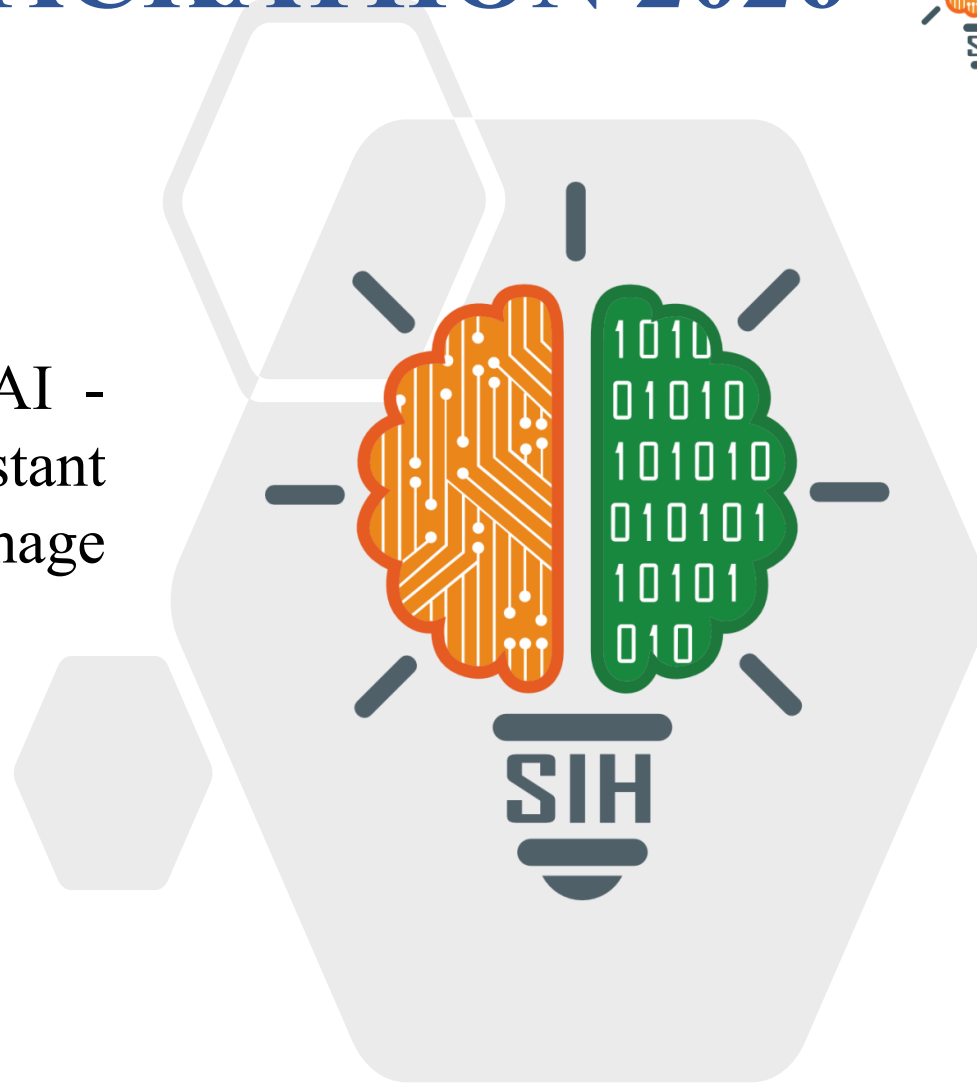


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- **Problem Statement ID** : SIH26167
- **Problem Statement Title** : SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme** : Space Technology
- **PS Category** : Software
- **Team Name** : Astra Nova





SatQuery AI — Agentic Multimodal Satellite Intelligence Assistant



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SatQuery AI — An Agentic Vision-Language Assistant for Optical + SAR Remote Sensing

Challenge We're Solving

Silo problem: Every RS AI tool does one task only, so an analyst juggles six of them to answer one question.

Expertise wall: Users must know sensor, bands, resolution and file format before they can ask anything.

Cloud blindness: Optical fails during floods and monsoon, exactly when answers matter. SAR sits unused because it needs specialists.

Our Solution — SatQuery AI

One conversational assistant over a registry of specialist models. Ask in plain English (or Any language), the system plans the pipeline, runs it, and returns a verified answer.

1. Agentic Orchestrator

LangGraph + Qwen2.5-7B function calling auto-selects, sequences and executes the right tools for the query and input pair.

2. Specialist Tool Registry

RS-VQA, referring segmentation, bi-temporal change detection and captioning, SAR water and built-up analytics.

3. Cross-Modal Fusion Engine

Co-registers optical/multispectral with SAR and fuses complementary evidence into a single answer.

4. Evidence Layer (never bare text)

Every reply ships a mask and GeoJSON overlay, a change map, area in km^2/ha , a confidence score and the full agent trace.

How It Addresses the Problem

Each mandated query maps to an automatic tool chain:

“Describe the land-cover and major objects.”

→ captioning + LULC segmentation + object detector, returning a narrative with a class-wise area table.

“Highlight the water body referred to.”

→ referring-expression segmentation, returning a mask on the map plus extent in hectares.

“What changed between these two dates, and where?”

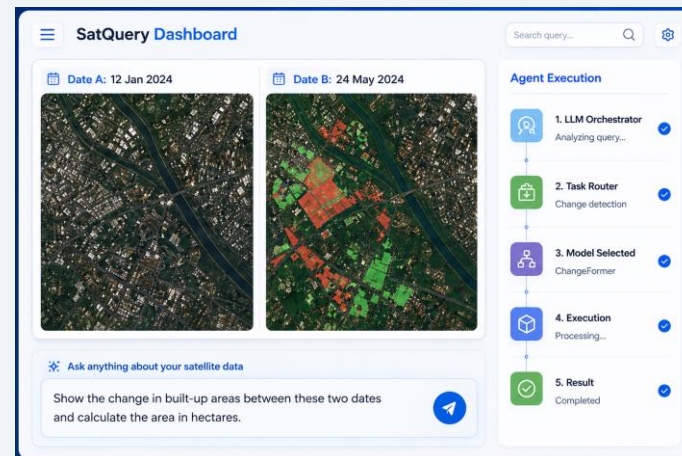
→ bi-temporal change VQA plus a spatial change map (the mandatory capability) and change captioning.

“Use optical and SAR together for built-up and water.”

→ co-registration, then SAR backscatter thresholding fused with optical indices into a two-class map.

“Has built-up area increased, decreased or stayed same?”

→ class-wise change quantification giving direction, Δkm^2 and percentage, computed from pixels.



Innovation & Uniqueness

1. SAR is first-class, not an afterthought.

Nearly every RS vision-language model (GeoChat, RS-LLaVA, EarthDial) is optical-only. We add a dedicated SAR chain: speckle filtering, Otsu backscatter water extraction, texture-based built-up, polarimetric decomposition. Answers survive cloud and night.

2. Grounded, not generative.

The LLM never invents a number. Areas come from mask pixel-count $\times$ ground sample distance, coordinates from the raster CRS. Below a confidence threshold the system abstains.

3. Glass-box agent.

The tool chain is streamed live and exportable, so an analyst can audit which model produced which claim.

4. Runs fully air-gapped.

All weights local; zero cloud API, zero recurring cost, deployable inside ISRO/NRSC networks.

5. Indian-data native.

Built on Bhoonidhi — NISAR S-SAR, EOS-04 C-SAR, Resourcesat LISS-III/AWiFS — plus Bhuvan thematic layers, not just Sentinel.

Beyond the Problem Statement

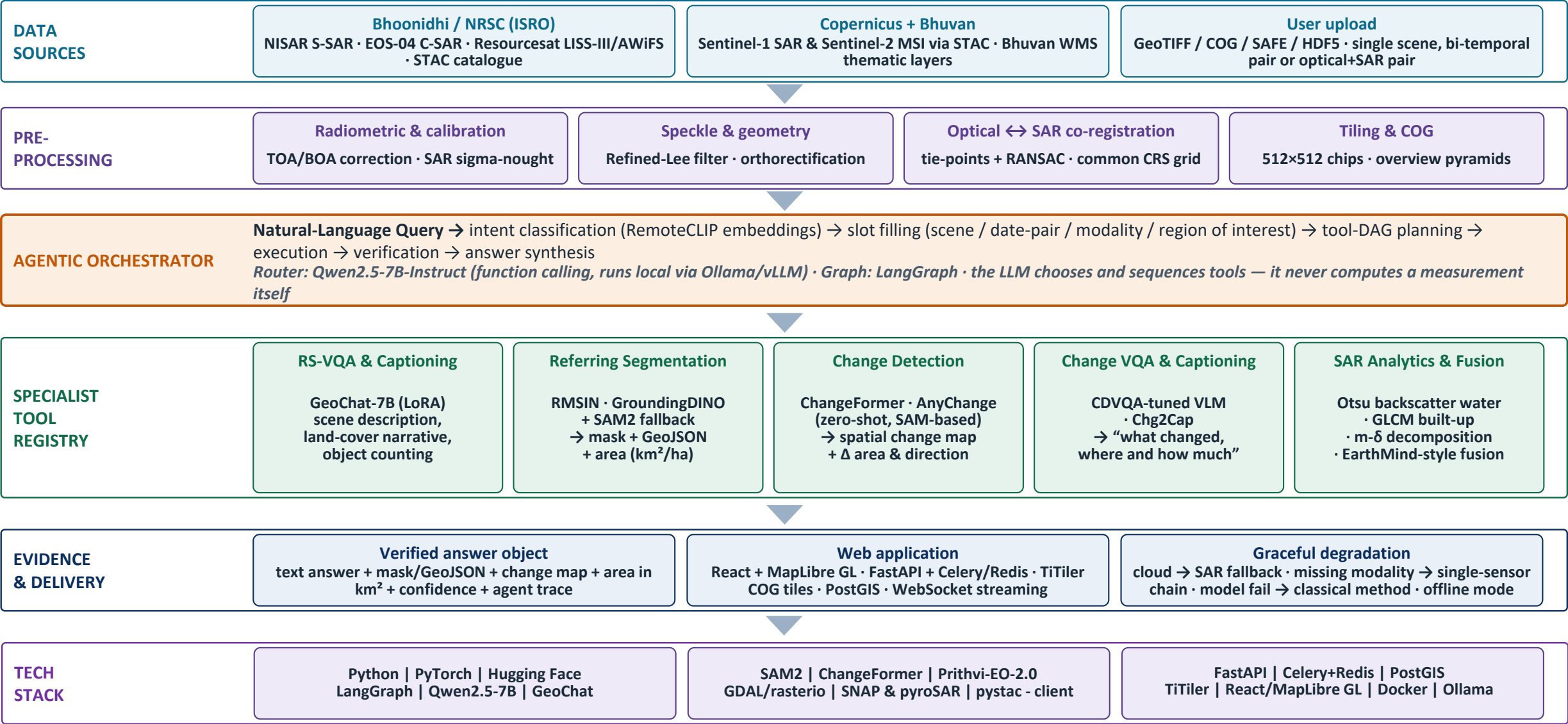
Multilingual querying in Hindi and regional languages for district-level users · **standing alerts** such as “tell me when built-up grows in this ward” · **batch district reports** from one query · **MCP tool exposure** so other ISRO systems can call the same specialist registry.



TECHNICAL APPROACH



System Architecture — from satellite archive to a grounded answer on the map





FEASIBILITY AND VIABILITY



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Technical Feasibility

No model trained from scratch. Every specialist is an open pretrained checkpoint — GeoChat-7B, RMSIN, ChangeFormer, SAM2, Prithvi-EO-2.0 (CC-BY-4.0). Only LoRA adaptation is needed.

Single-GPU build. 4-bit 7B router + specialist inference fits a 16–24 GB GPU; no cluster required for the prototype.

Data already exists and is free. Bhoonidhi open tier + Copernicus STAC for imagery; RSVQA, CDVQA, RRSIS-D, LEVIR-CD/CC and SEN12MS for training and evaluation.

Economic Viability

Fully open-source stack — no per-query LLM API cost and no vendor lock-in. Marginal cost per query on an on-prem node is effectively electricity only. One GPU node serves a departmental pilot; scaling is horizontal via Celery workers and cached COG tiles.

Operational & Legal Viability

Air-gapped deployment keeps sensitive imagery inside ISRO/NRSC networks — no data leaves the premises. Aligned with the Indian Space Policy 2023 and the National Geospatial Policy 2022. Model licences are checked per component (Apache-2.0 router, CC-BY-4.0 encoder) so the stack stays deployable.

Potential Challenges & Risks

1. Optical–SAR misregistration

Different geometry and terrain distortion make pixel-level fusion drift, corrupting every downstream mask.

2. Vision-language hallucination

VLMs confidently invent object counts and areas — unacceptable for an operational space product.

3. Domain gap

Public models are trained largely on US/European imagery; Indian landscapes, field sizes and settlement patterns differ.

4. Cloud cover & missing inputs

Monsoon optical scenes are unusable, and a required modality or date may simply be absent.

5. Latency on full scenes

A multi-GB scene through a chain of deep models is far too slow for interactive questioning.

6. Integration risk (36 hours)

Many moving parts; most hackathon failures happen at the seams, not inside a single model.

7. “Is this real space data or a demo set?”

Evaluators discount systems demonstrated on curated toy imagery rather than operational products.

Strategies for Overcoming Them

Start from terrain-corrected ARD

Bhoonidhi EOS-04/NISAR products are already ortho-normalised; we refine with tie-point + RANSAC matching and report registration RMSE with every fused answer.

Numbers come from pixels, never the LLM

Area = mask pixel count × ground sample distance; coordinates from the raster CRS. Below a confidence threshold the agent abstains and says so.

LoRA on Indian scenes + honest evaluation

Adapt on Bhoonidhi imagery and validate on a hand-labelled Indian eval set, reporting mIoU, F1 and VQA accuracy rather than demo screenshots.

SAR as the designed fallback

Cloud detected → automatic switch to the SAR chain; missing modality → single-sensor path; model failure → classical thresholding. The system degrades, it does not break.

Tile-level inference and caching

Run models on the requested tiles only, cache in Redis, stream partial results — COG lets the browser fetch kilobytes instead of gigabytes.

Contract-first, integrate early

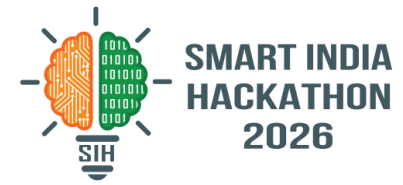
Every tool implements one typed interface behind a stub, so the end-to-end path runs in hour six and real models are swapped in one by one.

Provenance shown on screen

Every demo scene is pulled from Bhoonidhi or Copernicus with its product ID, sensor and acquisition date displayed beside the answer.



IMPACT AND BENEFITS



TARGET AUDIENCE — WHO BENEFITS AND HOW

ISRO / NRSC analysts

Turn a multi-tool GIS workflow into one sentence. Query the Bhoonidhi archive conversationally instead of scripting per-task pipelines.

Disaster management (NDMA, state cells)

Flood extent from SAR through cloud, day or night — inundated area in km² within minutes of a pass, not hours.

Agriculture & water resources

Crop-area and water-body change tracking for PMFBY assessment, reservoir monitoring and command-area planning.

Urban bodies & forestry

Built-up growth quantified between two dates for AMRUT/Smart Cities master-plan review and forest-cover change reporting.

BENEFITS OF THE SOLUTION

Social

Democratises satellite intelligence. A district officer or farmer-producer group gets an answer without learning remote sensing, widening who can act on ISRO data.

Economic

Zero recurring API cost on an open stack. Feeds a national EO market projected to reach roughly \$8 bn of India's \$44 bn space-economy target by 2033 (FICCI-EY, 2025).

Environmental

Faster detection of deforestation, encroachment on water bodies, and unplanned urban sprawl — with measured extents that stand up in a review.

Strategic

Indigenous, air-gapped capability on Indian sensors (NISAR, EOS-04, Resourcesat). Supports Atmanirbhar Bharat and the Indian Space Policy 2023.

How We Will Measure That It Actually Works — After Deployment, Not Just At Demo Time

Time-to-answer

Minutes from query to verified map layer, against the current multi-tool GIS baseline

Accuracy vs reference

mIoU / F1 on change and segmentation masks; area error against reference masks where available

Expert independence

Share of queries resolved with no remote-sensing specialist in the loop

Operational uptake

Queries per week per department and repeat usage after the pilot period ends



RESEARCH AND REFERENCES



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ISRO / Government Data & Policy

- Bhoonidhi — ISRO EO Data Hub (NRSC): bhoonidhi.nrsc.gov.in
- NISAR S-SAR product release, Bhoonidhi: bhoonidhi.nrsc.gov.in/NISAR/NisarProducts.html
- NISAR Data User Guide (ASF DAAC): nisar-docs.asf.alaska.edu
- Bhuvan Geoportal — WMS/WMTS thematic services: bhuvan.nrsc.gov.in
- MOSDAC (SAC) data download API: mosdac.gov.in · VEDAS vAPI: vedas.sac.gov.in
- Indian Space Policy 2023 · National Geospatial Policy 2022 · IN-SPACe: inspace.gov.in

Vision-Language & Foundation Models

- Kuckreja et al., “GeoChat: Grounded Large Vision-Language Model for Remote Sensing”, CVPR 2024 (arXiv:2311.15826)
- Shu et al., “EarthMind: Leveraging Cross-Sensor Data ... Unified Multimodal LLM”, 2025 (arXiv:2506.01667) — optical + SAR fusion
- Liu et al., “RemoteCLIP: A Vision Language Foundation Model for Remote Sensing”, IEEE TGRS 2024 (arXiv:2306.11029)
- Szwarcman et al., “Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation” (arXiv:2412.02732)
- Zhang et al., “RS5M and GeoRSCLIP: A Large-Scale Vision-Language Dataset and Model for Remote Sensing”, IEEE TGRS

Agentic Orchestration for Geospatial AI

- Xu et al., “RS-Agent: Automating Remote Sensing Tasks through Intelligent Agents” (arXiv:2406.07089)
- Liu et al., “Change-Agent: Towards Interactive Comprehensive Remote Sensing Change Interpretation and Analysis”, IEEE TGRS 2024
- Guo et al., “Remote Sensing ChatGPT: Solving Remote Sensing Tasks with ChatGPT and Visual Models”
- Danish et al., “GeoBench-VLM: Benchmarking Vision-Language Models for Geospatial Tasks” (arXiv:2411.19325)
- LangGraph documentation — stateful multi-actor agent graphs: langchain.com/langgraph

Datasets & Benchmarks

- RSVQA — visual question answering for remote sensing (arXiv:2003.07333)
- CDVQA — change-detection VQA (arXiv:2112.06343); CDQAG/VisTA (arXiv:2410.23828)
- RRSIS-D & RefSegRS — referring RS image segmentation (arXiv:2312.12470, 2306.08625)
- LEVIR-CD / LEVIR-CC — change detection & change captioning: justchenhao.github.io/LEVIR
- SEN12MS — Sentinel-1 SAR + Sentinel-2 optical fusion corpus (arXiv:1906.07789)
- DIOR-RSVG — visual grounding · RSICD, NWPU-Captions — RS captioning

Change Detection, Segmentation & SAR

- Zheng et al., “Segment Any Change” (AnyChange), NeurIPS 2024 (arXiv:2402.01188)
- Fang et al., “Changer: Feature Interaction is What You Need for Change Detection” (arXiv:2209.08290)
- Liu et al., “Rotated Multi-Scale Interaction Network for Referring RS Image Segmentation” (RMSIN), CVPR 2024
- Chang & Ghamisi, “Changes to Captions: An Attentive Network for RS Change Captioning” (Chg2Cap, arXiv:2304.01091)
- Review of multi-resolution SAR–optical image registration methods (arXiv:2502.01002)

Engineering Stack & Market Context

- Cloud-Optimized GeoTIFF specification: cogeo.org · STAC specification: stacspec.org
- TiTiler — dynamic COG tile server (Development Seed): developmentseed.org/titiler
- Copernicus Data Space Ecosystem APIs: documentation.dataspace.copernicus.eu
- MapLibre GL JS · GDAL/rasterio · ESA SNAP & pyroSAR — SAR preprocessing toolchain
- FICCI-EY, “Unlocking India's Space Economy”, March 2025: ficci.in/press_release_details/5033
- Geospatial World, “Geospatial Artha” Report 2021 — Indian geospatial economy sizing